

Introduction to PXDataset2 object

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Resources

Bioconductor packages tagged by “proteomics”, n=188 on 8th Apr2026: https://bioconductor.org/packages/release/BiocViews.html#___Proteomics

```
knitr::opts_chunk$set(echo = FALSE)
## install.package("BiocManager")

# library(BiocManager)
#
# # Bioconductor packages
# if (!requireNamespace("BiocManager", quietly = TRUE))
#   install.packages("BiocManager")
#
# BiocManager::install("tidyverse")
# BiocManager::install("factoextra")
# BiocManager::install("msdata")
# BiocManager::install("mzR")
# BiocManager::install("rhd5")
# BiocManager::install("rpx")
# BiocManager::install("MsCoreUtils")
# BiocManager::install("QFeatures")
# BiocManager::install("Spectra")
# BiocManager::install("ProtGenerics")
# BiocManager::install("PSMatch")
# BiocManager::install("pheatmap")
# BiocManager::install("limma")
# BiocManager::install("MSnID")
# BiocManager::install("rawrr")
# BiocManager::install("RforMassSpectrometry/SpectraVis")
#
# install.packages("tidyverse")
```

Review some points about MS data

- The continuous workflow of separating-separating-separating!
- Three dimensions: Retention time of analyser elution - Mass to charge (M/Z) - Intensity

Spectrum is the unit of mass spectrometry data

TIC (total ion chromatogram) plot: The intensities of all ions in each mass spectrometer scan (within a given time period) are summed and plotted as a function of time. *Base peak chromatogram plot:* The intensity of the base peak (the most intense ion in each mass spectrum) is plotted as a function of time (Note that the base peak should not be confused with the baseline, which represents the constant background signal of the

instrument). *XIC or EIC plot*: The intensity of a specific analyte at a chosen m/z value (either total or base peak intensity) is isolated and plotted as a function of time. Chromatogram plots [Group Activity Open](#) “GroupActivity_20Apr.html” file.

Getting data

Get to know the ProteomeXchange Consortium: Established to create globally coordinated, standardized pipelines for data submission and dissemination across major proteomics repositories, the Consortium promotes open data policies in proteomics.

proteomexchange.org

```
#Direct accessing data
library(rpx)
library(tidyr)
px <- PXDataset("PXD022816")
```

```
## Loading PXD022816 from cache.
```

```
pxget(px, grep("mzID", pxfiles(px))[1:3]) # It might take time for the local access
```

```
## Project PXD022816 files (32):
```

```
## [local] QEP2LC6_HeLa_50ng_251120_01-calib.mzID.gz
## [local] QEP2LC6_HeLa_50ng_251120_01-calib.mzML
## [remote] QEP2LC6_HeLa_50ng_251120_01.raw
## [local] QEP2LC6_HeLa_50ng_251120_02-calib.mzID.gz
## [local] QEP2LC6_HeLa_50ng_251120_02-calib.mzML
## [remote] QEP2LC6_HeLa_50ng_251120_02.raw
## [local] QEP2LC6_HeLa_50ng_251120_03-calib.mzID.gz
## [local] QEP2LC6_HeLa_50ng_251120_03-calib.mzML
## [remote] QEP2LC6_HeLa_50ng_251120_03.raw
## [remote] QEP2LC6_HeLa_50ng_251120_04-calib.mzID.gz
## ...
```

```
## Loading QEP2LC6_HeLa_50ng_251120_01-calib.mzID.gz from cache.
```

```
## Loading QEP2LC6_HeLa_50ng_251120_02-calib.mzID.gz from cache.
```

```
## Loading QEP2LC6_HeLa_50ng_251120_03-calib.mzID.gz from cache.
```

```
## [1] "/Users/nvd448/Library/Caches/org.R-project.R/R/rpx/b8b73444262_QEP2LC6_HeLa_50ng_251120_01-calib.mzID.gz"
```

```
## [2] "/Users/nvd448/Library/Caches/org.R-project.R/R/rpx/b8b77662319a_QEP2LC6_HeLa_50ng_251120_02-calib.mzID.gz"
```

```
## [3] "/Users/nvd448/Library/Caches/org.R-project.R/R/rpx/b8b728a6b22e_QEP2LC6_HeLa_50ng_251120_03-calib.mzID.gz"
```

```
pxget(px, grep("mzML", pxfiles(px))[1:3]) # It might take time for the local access
```

```
## Project PXD022816 files (32):
```

```
## [local] QEP2LC6_HeLa_50ng_251120_01-calib.mzID.gz
## [local] QEP2LC6_HeLa_50ng_251120_01-calib.mzML
## [remote] QEP2LC6_HeLa_50ng_251120_01.raw
## [local] QEP2LC6_HeLa_50ng_251120_02-calib.mzID.gz
## [local] QEP2LC6_HeLa_50ng_251120_02-calib.mzML
## [remote] QEP2LC6_HeLa_50ng_251120_02.raw
## [local] QEP2LC6_HeLa_50ng_251120_03-calib.mzID.gz
## [local] QEP2LC6_HeLa_50ng_251120_03-calib.mzML
## [remote] QEP2LC6_HeLa_50ng_251120_03.raw
## [remote] QEP2LC6_HeLa_50ng_251120_04-calib.mzID.gz
## ...
```

```

## Loading QEP2LC6_HeLa_50ng_251120_01-calib.mzML from cache.
## Loading QEP2LC6_HeLa_50ng_251120_02-calib.mzML from cache.
## Loading QEP2LC6_HeLa_50ng_251120_03-calib.mzML from cache.
## [1] "/Users/nvd448/Library/Caches/org.R-project.R/R/rpx/b8b757fbfedb_QEP2LC6_HeLa_50ng_251120_01-calib.mzML"
## [2] "/Users/nvd448/Library/Caches/org.R-project.R/R/rpx/aaa44b473b55_QEP2LC6_HeLa_50ng_251120_02-calib.mzML"
## [3] "/Users/nvd448/Library/Caches/org.R-project.R/R/rpx/aaa474750230_QEP2LC6_HeLa_50ng_251120_03-calib.mzML"
(px <- PXDataset("PXD000001"))

## Loading PXD000001 from cache.
## Project PXD000001 with 11 files
##
## Resource ID BFC9 in cache in /Users/nvd448/Library/Caches/org.R-project.R/R/rpx.
## [1] 'F063721.dat' ... [11] 'erwinia_carotovora.fasta'
## Use 'pxfiles(.)' to see all files.
(pxfiles(px)) # file names of the object

## Project PXD000001 files (11):
## [local] F063721.dat
## [remote] F063721.dat-mztab.txt
## [remote] PRIDE_Exp_Complete_Ac_22134.xml.gz
## [remote] PRIDE_Exp_mzData_Ac_22134.xml.gz
## [remote] PXD000001_mztab.txt
## [local] README.txt
## [local] TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzML
## [remote] TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzXML
## [remote] TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01.mzXML
## [remote] TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01.raw
## ...
## [1] "F063721.dat"
## [2] "F063721.dat-mztab.txt"
## [3] "PRIDE_Exp_Complete_Ac_22134.xml.gz"
## [4] "PRIDE_Exp_mzData_Ac_22134.xml.gz"
## [5] "PXD000001_mztab.txt"
## [6] "README.txt"
## [7] "TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzML"
## [8] "TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzXML"
## [9] "TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01.mzXML"
## [10] "TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01.raw"
## [11] "erwinia_carotovora.fasta"
fn <- "TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzML"
mzf <- pxget(px, fn)

## Loading TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzML from cache.
pntax(px) # taxonomic name

## [1] "Erwinia carotovora"
pxref(px) # bibliography ref

## [1] "Gatto L, Christoforou A; Using R and Bioconductor for proteomics data analysis., Biochim Biophys Res Commun 2014;475:1-6."

```

```

pxget(px, "erwinia_carotovora.fasta")

## Loading erwinia_carotovora.fasta from cache.
## [1] "/Users/nvd448/Library/Caches/org.R-project.R/R/rpx/aaa45e66a84_erwinia_carotovora.fasta"
pxget(px, "README.txt") # a vector defining the files to be found, try to find it on your device.

## Loading README.txt from cache.
## [1] "/Users/nvd448/Library/Caches/org.R-project.R/R/rpx/aaa45a5b0322_README.txt"
# Some files are accessible remotely and some of them from local caching mechanism.
pxfiles(px) %>% str() # To check

## Project PXD000001 files (11):
## [local] F063721.dat
## [remote] F063721.dat-mztab.txt
## [remote] PRIDE_Exp_Complete_Ac_22134.xml.gz
## [remote] PRIDE_Exp_mzData_Ac_22134.xml.gz
## [remote] PXD000001-mztab.txt
## [local] README.txt
## [local] TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzML
## [remote] TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzXML
## [remote] TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01.mzXML
## [remote] TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01.raw
## ...

## chr [1:11] "F063721.dat" "F063721.dat-mztab.txt" ...
(f <- pxget(px, "TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzML"))

## Loading TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzML from cache.
## [1] "/Users/nvd448/Library/Caches/org.R-project.R/R/rpx/aaa481778b6_TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzML"
basename(f)

## [1] "aaa481778b6_TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzML"

```

MS example data: msdata

```

# Various Mass Spectrometry raw data example files for proteomics and metabolomics
library(msdata)
proteomics()

## [1] "MRM-standmix-5.mzML.gz"
## [2] "MS3TMT10_01022016_32917-33481.mzML.gz"
## [3] "MS3TMT11.mzML"
## [4] "TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzML.gz"
## [5] "TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01.mzML.gz"

ident()

## [1] "TMT_Erwinia_1uLSike_Top10HCD_isol2_45stepped_60min_01-20141210.mzid"

quant()

## [1] "cptac_a_b_peptides.txt"

```

Quiz 1 (GA_4)

Visit gosocrative.com and enter room name FIN2026

Raw MS data

- We need to convert binary-vendor-specific formats to open formats such as mzML, mzXML.
- mzML is the most recent format.
- proteowizard msconvert: <https://proteowizard.sourceforge.io/> or search in Google msconvert GUI
- ThermoRawFileParser: <https://github.com/compomics/ThermoRawFileParser>
- To check all MS-related data format: https://en.wikipedia.org/wiki/Mass_spectrometry_data_format

MS data from R prospective

Three dimensions How it looks in R